

SERVICE MANUAL

GE Healthcare Coils USA Instruments

Document 2413107-3

Revision 3

GE Signa[®] EXCITE[™] 3.0T Premier III Phased Array CTL Spine Coil (3.0T HD CTL Array)

GE Catalog Part Number: M1385AW



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Rev 3
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VÝSTRAHA (CS)	Tento provozní návod existuje pouze v anglickém jazyce. - V případě, že externí služba zákazníkům potřebuje návod v jiném jazyce, je zajištění překladu do odpovídajícího jazyka úkolem zákazníka. - Nesnažte se o údržbu tohoto zařízení, aniž byste si přečetli tento provozní návod a pochopili jeho obsah. - V případě nedodržování této výstrahy může dojít k poranění pracovníka prodejního servisu, obslužného personálu nebo pacientů vlivem elektrického proudu, respektive vlivem mechanických či jiných rizik.
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ĮSPĖJIMAS (LT)	Šis eksploatavimo vadovas yra tik anglų kalba. - Jei kliento paslaugų tiekėjas reikalauja vadovo kita kalba – ne anglų, suteikti vertimo paslaugas privalo klientas. - Nemėginkite atlikti įrangos techninės priežiūros, jei neperskaitėte ar nesupratote šio eksploatavimo vadovo. - Jei nepaisysite šio įspėjimo, galimi paslaugų tiekėjo, operatoriaus ar paciento sužalojimai dėl elektros šoko, mechaninių ar kitų pavojų.

ADVARSEL (NO)	Denne servicehåndboken finnes bare på engelsk. - Hvis kundens serviceleverandør har bruk for et annet språk, er det kundens ansvar å sørge for oversettelse. - Ikke forsøk å reparere utstyret uten at denne servicehåndboken er lest og forstått. - Manglende hensyn til denne advarselen kan føre til at serviceleverandøren, operatøren eller pasienten skades på grunn av elektrisk støt, mekaniske eller andre farer.
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AVISO (PT-BR)	Este manual de assistência técnica encontra-se disponível unicamente em inglês. - Se outro serviço de assistência técnica solicitar a tradução deste manual, caberá ao cliente fornecer os serviços de tradução. - Não tente reparar o equipamento sem ter consultado e compreendido este manual de assistência técnica. - A não observância deste aviso pode ocasionar ferimentos no técnico, operador ou paciente decorrentes de choques elétricos, mecânicos ou outros.
ATENÇÃO (PT-PT)	Este manual de assistência técnica só se encontra disponível em inglês. - Se qualquer outro serviço de assistência técnica solicitar este manual noutro idioma, é da responsabilidade do cliente fornecer os serviços de tradução. - Não tente reparar o equipamento sem ter consultado e compreendido este manual de assistência técnica. - O não cumprimento deste aviso pode colocar em perigo a segurança do técnico, do operador ou do paciente devido a choques elétricos, mecânicos ou outros.
ATENȚIE (RO)	Acest manual de service este disponibil doar în limba engleză. - Dacă un furnizor de servicii pentru clienți necesită o altă limbă decât cea engleză, este de datoria clientului să furnizeze o traducere. - Nu încercați să reparați echipamentul decât ulterior consultării și înțelegerii acestui manual de service. - Ignorarea acestui avertisment ar putea duce la rănirea depanatorului, operatorului sau pacientului în urma pericolelor de electrocutare, mecanice sau de altă natură.
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UPOZORENJE (SR)	Ovo servisno uputstvo je dostupno samo na engleskom jeziku. - Ako klijentov serviser zahteva neki drugi jezik, klijent je dužan da obezbedi prevodilačke usluge. - Ne pokušavajte da opravite uređaj ako niste pročitali i razumeli ovo servisno uputstvo. - Zanemarivanje ovog upozorenja može dovesti do povređivanja serviser, rukovaoca ili pacijenta usled strujnog udara ili mehaničkih i drugih opasnosti.

UPOZORNENIE (SK)	Tento návod na obsluhu je k dispozícii len v angličtine. • Ak zákazníkovi poskytovateľ služieb vyžaduje iný jazyk ako angličtinu, poskytnutie prekladateľských služieb je zodpovednosťou zákazníka. • Nepokúšajte sa o obsluhu zariadenia, kým si neprečítate návod na obsluhu a neporozumiete mu. • Zanedbanie tohto upozornenia môže spôsobiť zranenie poskytovateľa služieb, obsluhujúcej osoby alebo pacienta elektrickým prúdom, mechanické alebo iné ohrozenie.
ATENCION (ES)	Este manual de servicio sólo existe en inglés. • Si el encargado de mantenimiento de un cliente necesita un idioma que no sea el inglés, el cliente deberá encargarse de la traducción del manual. • No se deberá dar servicio técnico al equipo, sin haber consultado y comprendido este manual de servicio. • La no observancia del presente aviso puede dar lugar a que el proveedor de servicios, el operador o el paciente sufran lesiones provocadas por causas eléctricas, mecánicas o de otra naturaleza.
VARNING (SV)	Den här servicehandboken finns bara tillgänglig på engelska. . • Om en kunds servicetekniker har behov av ett annat språk än engelska, ansvarar kunden för att tillhandahålla översättningstjänster. • Försök inte utföra service på utrustningen om du inte har läst och förstår den här servicehandboken. • Om du inte tar hänsyn till den här varningen kan det resultera i skador på serviceteknikern, operatören eller patienten till följd av elektriska stötar, mekaniska faror eller andra faror.
DİKKAT (TR)	Bu servis kılavuzunun sadece ingilizcesi mevcuttur. • Eğer müşteri teknisyeni bu kılavuzu ingilizce dışında bir başka lisandan talep ederse, bunu tercüme ettirmek müşteriye düşer. • Servis kılavuzunu okuyup anlamadan ekipmanlara müdahale etmeyiniz. • Bu uyarıya uyulmaması, elektrik, mekanik veya diğer tehlikelerden dolayı teknisyen, operatör veya hastanın yaralanmasına yol açabilir.

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SECTION 1 – INTRODUCTION

1-1 Product Identification and Shipping List

To identify the Premier III Phased Array CTL Spine Coil, refer to the serial number label for GE part number 2413107. *Figure 1a* is a photograph of the coil and *Figure 1b* is of the serial number label.

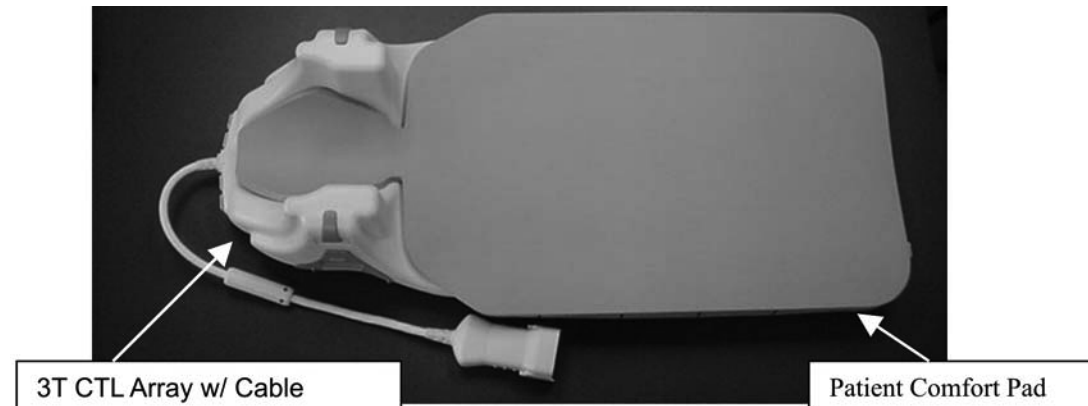


Figure 1a: 3.0T 8 Channel CTL Spine Coil

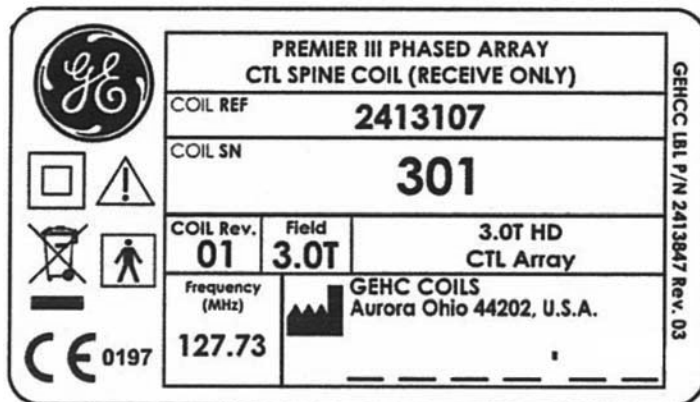


Figure 1b: Example Serial Number for an 8 Channel CTL Spine Coil

SHIPPING LIST – TABLE 1-1

Box #	Part Name	GE Part #	Qty
1	Premier III Phased Array CTL Spine Coil w/ Cable	2413107	1
1	Patient Comfort Pad	2413401	1
1	Ear Pads	2414423	2
1	Disposable Ear Pad Covers	U1-150536	500
1	Operator Guide Packet	2422232-500	1
1	Service Manual	2413107-3	1

1-2 Compatibility

This coil is compatible with the Signa[®] Excite 3.0T system.

1-3 Related Documentation

Operator's Guide, GE Part Number 2422232.

Signa[®] EXCITE 3T 11.x Service Methods CD, 2389160-1.

Signa[®] EXCITE 3T 12.x Service Methods CD, 5133330-1.

1-4 Environmental Requirements

Store the coil in an air-conditioned scan room or equipment room.

Environmental Requirements – TABLE 1-4

	Transport and Storage	Operating Conditions
Barometric Pressure	860hPa to 1060hPa	As established by the GE recommended MR suite conditions
Humidity	20% to 95% non-condensing	45% to 75% non-condensing
Temperature	-15° C to 55° C (5° F to 131° F)	15° C to 35° C (59° F to 95° F)

1-5 Theory of Operation

The Premier III Phased Array CTL Spine Coil is a receive-only coil with a total of 12 coil elements, 6 loop elements and 6 saddle elements (see *Figure 2*). It is designed to give optimum signal to noise ratio and uniform coverage of the spine anatomy, including the cervical, thoracic and lumbar regions. It can be operated in 12 different modes. An electronic multiplexer inside the coil, controlled by the MR system, selects the coil elements for each mode (see *Figure 3*). The unselected coil elements are turned off by the DC currents provided by the MR system. *Table 1-5* shows the selected elements and clinic use of each mode.

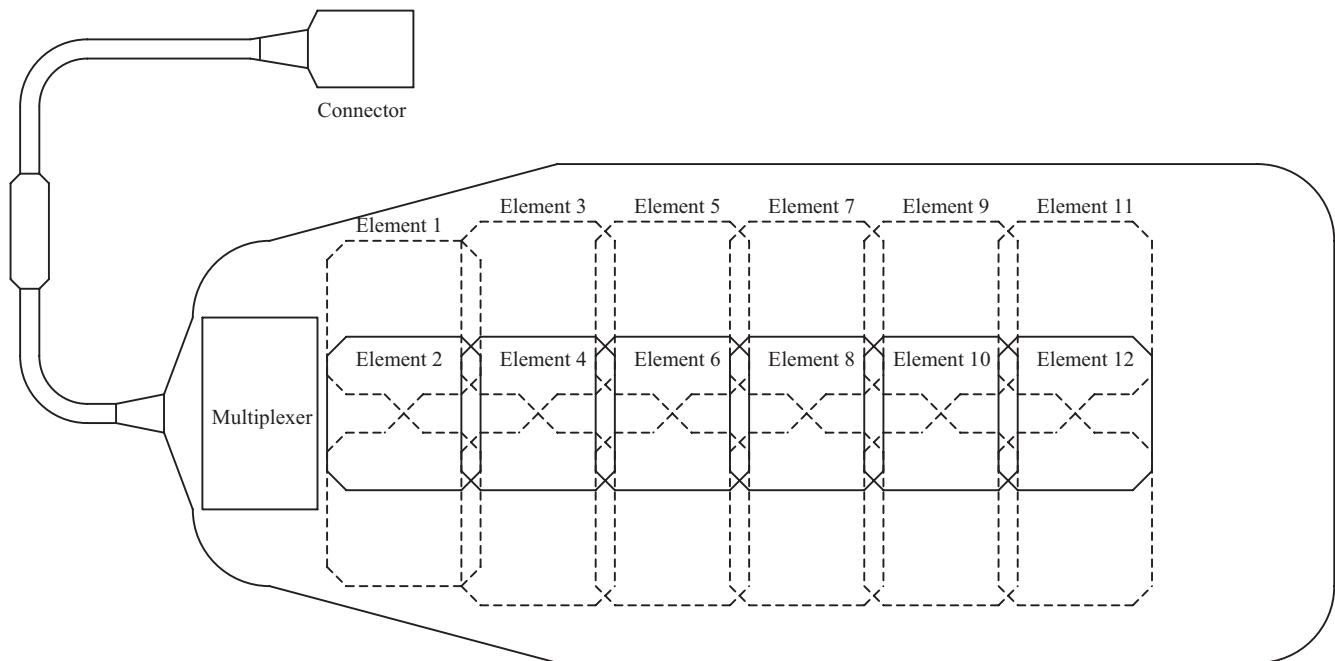


Figure 2: The Layout of the Elements

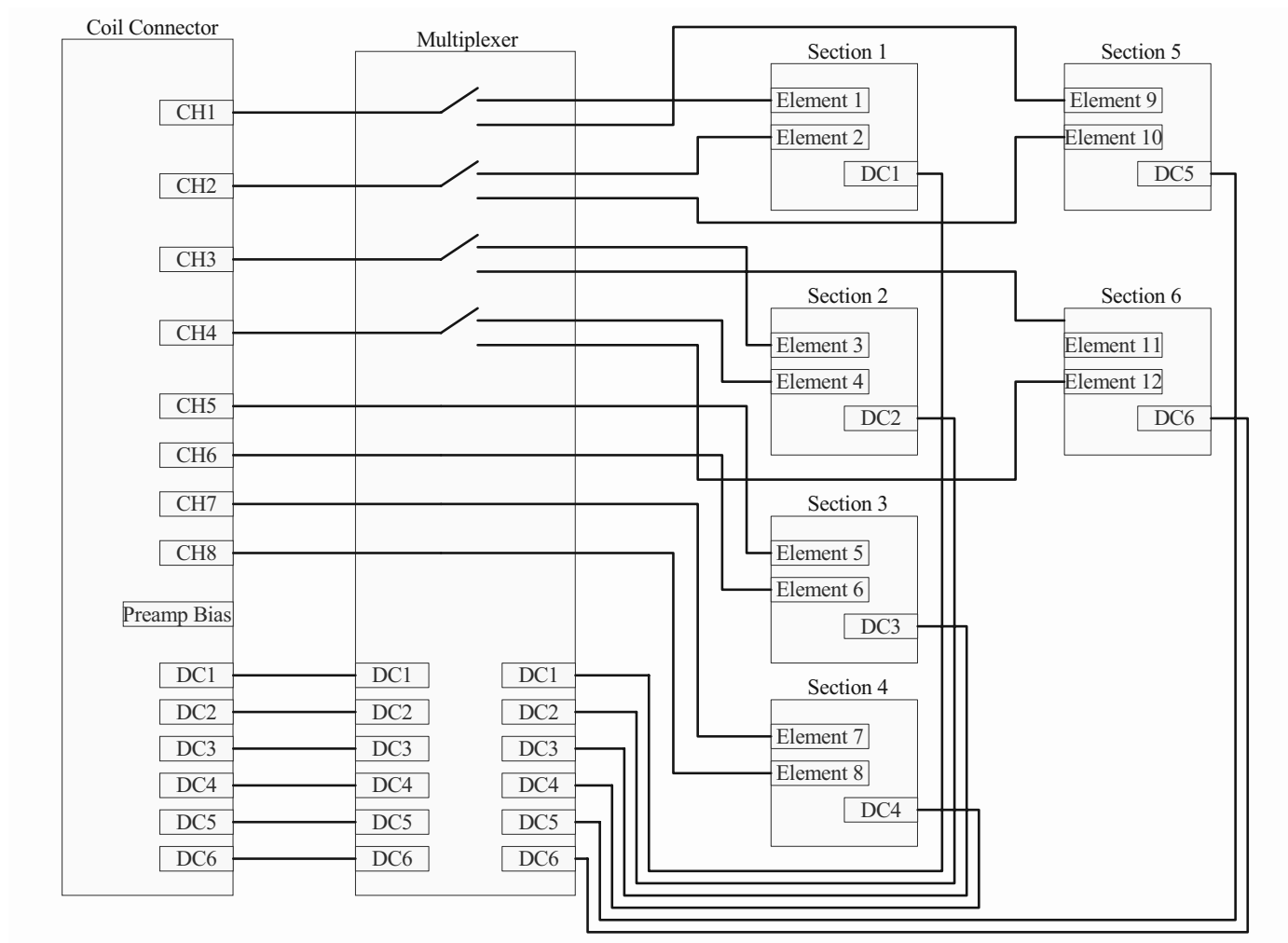


Figure 3: Coil Diagram

Modes of Operation – TABLE 1-5

Coil Mode (for 11.0 system)	Coil Mode (for 12.0 system)	Selected Elements	Clinical Use
8USCTLTOP	HDCTL TOP	1, 2, 3, 4, 5, 6, 7, 8	Cervical and thoracic spine
8USCTLMID	HDCTL MID	3, 4, 5, 6, 7, 8, 9, 10	Thoracic and lumbar spine
8USCTLBOT	HDCTL BOT	5, 6, 7, 8, 9, 10, 11, 12	Thoracic and lumbar spine
8USCTL123	HDCTL 123	1, 2, 3, 4, 5, 6	Cervical and thoracic spine
8USCTL234	HDCTL 234	3, 4, 5, 6, 7, 8	Thoracic spine
8USCTL345	HDCTL 345	5, 6, 7, 8, 9, 10	Thoracic and lumbar spine
8USCTL456	HDCTL 456	7, 8, 9, 10, 11, 12	Thoracic and lumbar spine
8USCTL12	HDCTL 12	1, 2, 3, 4	Cervical spine
8USCTL23	HDCTL 23	3, 4, 5, 6	Thoracic spine
8USCTL34	HDCTL 34	5, 6, 7, 8	Thoracic spine
8USCTL45	HDCTL 45	7, 8, 9, 10	Thoracic and lumbar spine
8USCTL56	HDCTL 56	9, 10, 11, 12	Lumbar spine

SECTION 2 – SETUP AND CALIBRATION

2-1 Coil Installation

2-1-1 Special Install Notes

None.

2-1-2 Configuration

This coil uses the following configuration names in 11.0 system:

8USCTLTOP
8USCTL MID
8USCTLBOT
8USCTL123
8USCTL234
8USCTL345
8USCTL456
8USCTL12
8USCTL23
8USCTL34
8USCTL45
8USCTL56
8USCTL123service
8USCTL456service
8USCTL12service
8USCTL34service
8USCTL56service

This coil uses the following configuration names in 12.0 system:

HDCTL TOP
HDCTL MID
HDCTL BOT
HDCTL 123
HDCTL 234
HDCTL 345
HDCTL 456
HDCTL 12
HDCTL 23
HDCTL 34
HDCTL 45
HDCTL 56
HDCTL 123serv
HDCTL 456serv
HDCTL 12serv
HDCTL 34serv
HDCTL 56serv

Add the coil configuration names using the Configuration File Manager. Refer to: Service Methods CD; Software; System Level Software Utilities; Configuration File Manager.

The following tables provide additional information about the coil. *Table 2-1-1* shows the relation between coil modes, receive channels and selected coil elements. *Table 2-1-2* shows the relation between coil modes and MC bias.

Coil modes, Receive channels and selected coil elements – Table 2-1-1

Coil Mode		Receive Channel							
Coil Mode for 11.0 System	Coil Mode for 12.0 System	CH 1	CH 2	CH 3	CH 4	CH 5	CH 6	CH 7	CH 8
8USCTLTOP	HDCTL TOP	E1	E2	E3	E4	E5	E6	E7	E8
8USCTLMID	HDCTL MID	E9	E10	E3	E4	E5	E6	E7	E8
8USCTLBOT	HDCTL BOT	E9	E10	E11	E12	E5	E6	E7	E8
8USCTL123	HDCTL 123	E1	E2	E3	E4	E5	E6		
8USCTL234	HDCTL 234			E3	E4	E5	E6	E7	E8
8USCTL345	HDCTL 345	E9	E10			E5	E6	E7	E8
8USCTL456	HDCTL 456	E9	E10	E11	E12			E7	E8
8USCTL12	HDCTL 12	E1	E2	E3	E4				
8USCTL23	HDCTL 23			E3	E4	E5	E6		
8USCTL34	HDCTL 34					E5	E6	E7	E8
8USCTL45	HDCTL 45	E9	E10					E7	E8
8USCTL56	HDCTL 56	E9	E10	E11	E12				
8USCTL123 service	HDCTL 123serv	E1	E2	E3	E4	E5	E6		
8USCTL456 service	HDCTL 456serv	E9	E10	E11	E12			E7	E8
8USCTL12 service	HDCTL 12serv	E1	E2	E3	E4				
8USCTL34 service	HDCTL 34serv					E5	E6	E7	E8
8USCTL45 service	HDCTL 56serv	E9	E10	E11	E12				

Coil modes and MC bias – Table 2-1-2

Coil Mode		MC Bias							
Coil Mode for 11.0 System	Coil Mode for 12.0 System	MC1	MC2	MC3	MC4	MC5	MC6	MC7	MC8
8USCTLTOP	HDCTL TOP	Toggle	Toggle	Toggle	Toggle			Not used	Not used
8USCTLMID	HDCTL MID		Toggle	Toggle	Toggle	Toggle		Not used	Not used
8USCTLBOT	HDCTL BOT			Toggle	Toggle	Toggle	Toggle	Not used	Not used
8USCTL123	HDCTL 123	Toggle	Toggle	Toggle				Not used	Not used
8USCTL234	HDCTL 234		Toggle	Toggle	Toggle			Not used	Not used
8USCTL345	HDCTL 345			Toggle	Toggle	Toggle		Not used	Not used
8USCTL456	HDCTL 456				Toggle	Toggle	Toggle	Not used	Not used
8USCTL12	HDCTL 12	Toggle	Toggle					Not used	Not used
8USCTL23	HDCTL 23		Toggle	Toggle				Not used	Not used
8USCTL34	HDCTL 34			Toggle	Toggle			Not used	Not used
8USCTL45	HDCTL 45				Toggle	Toggle		Not used	Not used
8USCTL56	HDCTL 56					Toggle	Toggle	Not used	Not used
8USCTL123 service	HDCTL 123serv	Toggle	Toggle	Toggle				Not used	Not used
8USCTL456 service	HDCTL 456serv				Toggle	Toggle	Toggle	Not used	Not used
8USCTL12 service	HDCTL 12serv	Toggle	Toggle					Not used	Not used
8USCTL134 service	HDCTL 34serv			Toggle	Toggle			Not used	Not used
8USCTL156 service	HDCTL 56serv					Toggle	Toggle	Not used	Not used

The CTL coil uses +7V for forward bias. The blank blocks in the above table means the MC bias is un-toggled.

2-2 Installation Functional Checks

Perform Section 3-2 Coil Imaging Performance Verification.

2-3 Periodic Quality Assurance Check

On a periodic basis, such as during planned maintenance, perform the quality assurance checks as outlined below to ensure the coil is operating properly.

1. Check external cable for cracks or cuts.
2. Perform Section 3-2 Coil Imaging Performance Verification.

SECTION 3 – FUNCTIONAL CHECKS

3-1 Scanner Verification

Perform system level Signal to Noise Check. Refer to Service Methods CD; System Level Procedures; Functional Checks; Signal to Noise Check.

3-2 Coil Imaging Performance Verification

NOTE: Legacy coils were shipped with a coil specified phantoms. New coils do not ship with phantoms; phantoms come in a unified phantom set with the MR system.

3-2-1 Tools Required

TOOLS REQUIRED – TABLE 3-1

Description	Qty	Effectivity	GE Part #
CT Phantom	1	Legacy	2381682-9
TL Phantom	2	Legacy	2381682-10
TL Unified Phantom, SiOil	2	Unified	5343347-2
Large Cylindrical unified phantom, SiOil	1	Unified	5342679-2

3-2-2 Data Collection

3-2-2-1 Legacy Phantom Procedure

Step 1 – Place the CTL coil on the patient cradle. Note the serial number of the coil.

Step 2 – Position the phantoms on the coil as shown in *Figure 4*.



Figure 4: Positioning the coil and phantoms

Step 3 – Landmark the coil at the raised mark closest to the head section of the coil (see *Figure 5a and 5b*).



Figure 5a

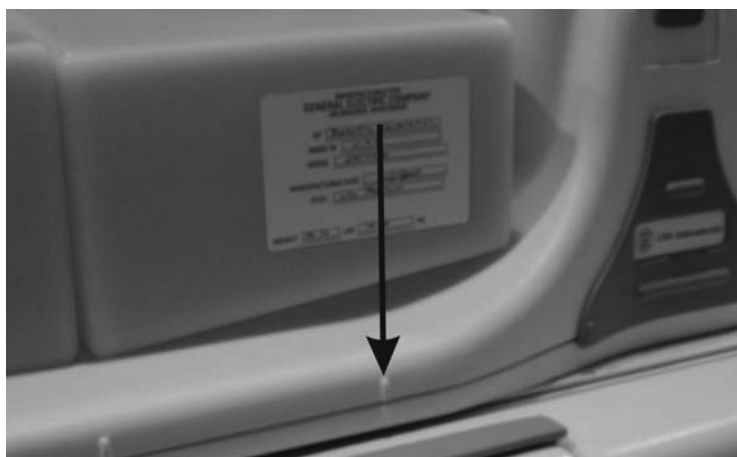


Figure 5b: Mark closest to the head section of the coil

3-2-2-2 Unified Phantom Procedure

Visual	Description
 Figure 6	Place the coil on the patient table and latch it to the table (<i>Figure 6</i>).
 Figure 7  Figure 8	Place the TL unified phantoms on the center of the TL section of the coil as shown in the <i>Figure 7</i>. Align the phantoms to the center in S/I and R/L directions as shown in <i>Figure 7</i> and <i>Figure 8</i>.



Figure 9

Place the large cylindrical unified phantom on the CT section as shown in *Figure 9*.



Figure 10

Adjust the large cylindrical unified phantom such that it is in complete contact with the TL unified phantom (*Figure 10*).

Adjust TL unified phantoms such that both the phantoms are in complete contact with each other as shown in *Figure 11*.

Connect the coil to respective port of the system LPCA.



Figure 11



Figure 12

Landmark the coil on the mark (*Figure 12*) and advance to scan.

Refer to Section 3-2-2-3 of this document to run scan with the Unified Phantom Set.

3-2-2-3 Steps to run MCQA

SETUP AND STARTING Multi-Coil Quality Assurance (MCQA) tool USING LEGACY PHANTOM SET

WARNING: All RF coil related tests must be run on a system that is well-calibrated and passing all system tests, especially white pixel, correlated noise, and MCR (Multi-Coil-Receive) tool.

1. If possible, run the Multi-Coil Chain Receive (MCRv) tool to validate receive chain function before running the MCQA tool. If the MCRv tool has missing or failed results, the MCQA tool will detect it and a pop-up will appear, asking if you want to continue.
2. From the Common Service Desktop (CSD) select [Image Quality], *Multi-Coil QA Tool* and Click here to start this tool. The Coil Log Viewer will open as shown in Figure 13.

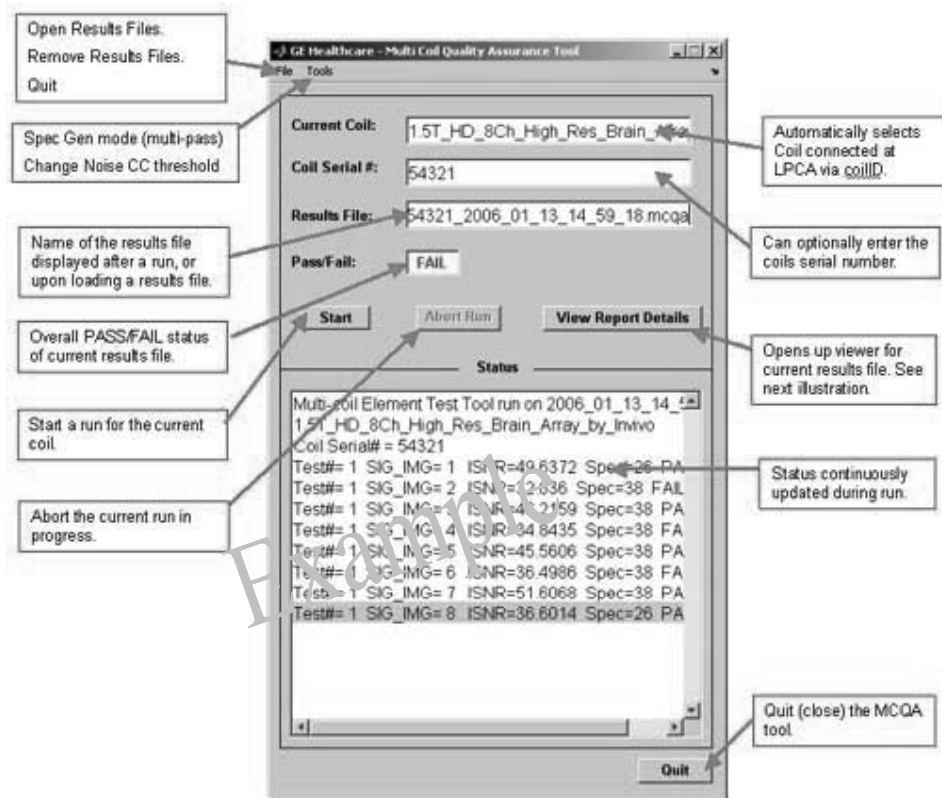


Figure 13: MCQA Starting window

3. The current coil field will be automatically filled in, based on the Coil ID of the coil connected to the LPCA. The names listed will be the physical coil names for each coil.

NOTE: Coil names are specific to the coil connector. Legacy coils should not have reference to the connector, but any new coil with the Bendix connector will have an HDx/HDe identifier in the coil name. During download, the tool should detect if the wrong coil was selected, and give an error. The user will then have the ability to pick again.

4. Enter the *Coil Serial #*: into the provided field.
5. From the MCQA Tool GUI, press the [Start] button to begin the test.
6. A warning dialog (*Figure 14*) will be shown, asking the user to verify the coil/phantom set-up and position. Ensure the phantom and coil setup is proper and select [Yes] to continue.
 - a. The status window of the MCQA Tool GUI will continuously update to give information on what the tool is doing at any point in time.
 - b. A time bar will appear, showing approximate total test time, elapsed time and percent complete, as shown in *Figure 15*.

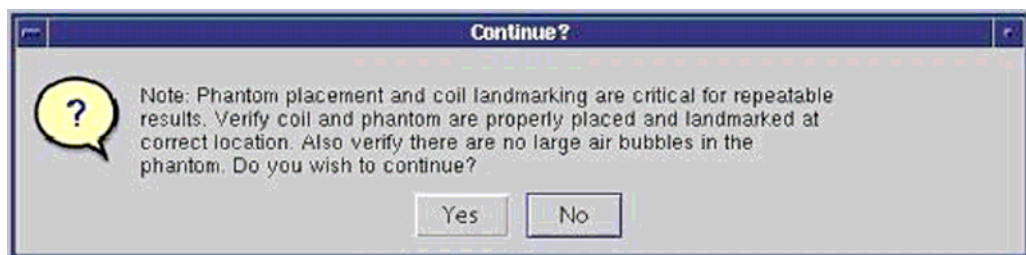


Figure 14: Phantom/Coil, Landmark Warning Popup

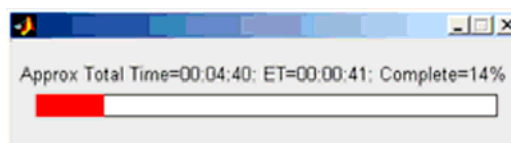


Figure 15: Status and Time Bar

7. Once the test has completed, the name of the Results file will be listed, along with the Pass/Fail status of the test.
8. Record the Measured ISNR and the LSL for each channel displayed in the Status Window on a separate sheet. (See *Table 4.1.3-1*)

Table 4.1.3- 1: Data Collection Table Example

Si No	Test No.	Element No.	LSL	ISNR

9. Select the [QUIT] button to exit out of the MCQA Tool.
10. Remove the coil and phantom from the system.

Setup and Starting Multi-coil QA (MCQA) Tool USING unified phantom KIT

All RF coil related tests must be run on a system that is well-calibrated and passing all system tests. The system should have passed "Install In Spec" (IIS), especially white pixel, correlated noise, and MCR (Multi-Coil-Receive) tool.

1. From Common Service Desktop (CSD) go to Service Browser and select [Image Quality], Multi-Coil QA Tool and then Click here to start this tool as shown in *Figure 16*.



Figure 16

2. Upon start-up, the coil that is connected to the system gets selected in the *Current Coil* field as shown in *Figure 17*.

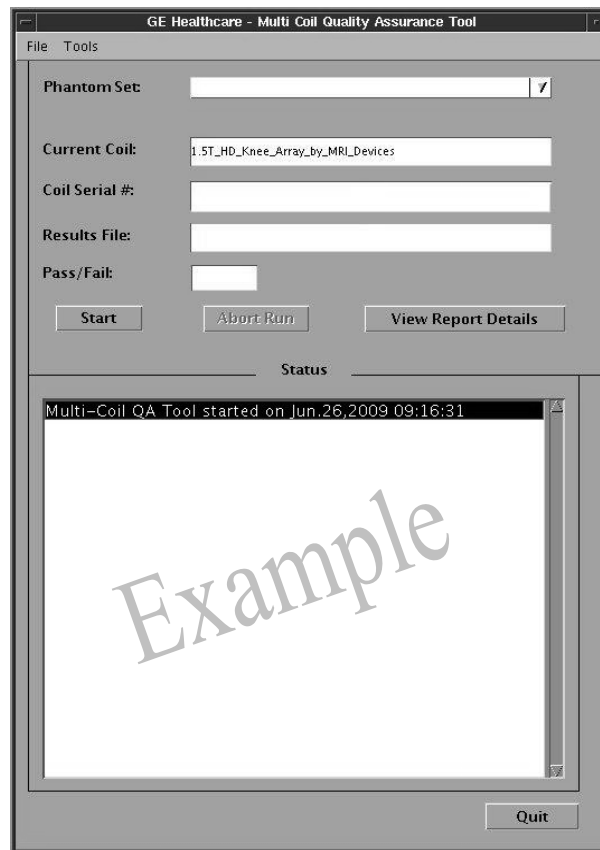


Figure 17

3. A pop-up displays requesting you to, **Please select correct Phantom Set** as shown in *Figure 18* below. Click on [OK] to continue.



Figure 18

NOTE: For Optima MR450w, the Unified Phantom Kit is the **ONLY** available phantom set and will be automatically selected in the [Phantom Set] field of the tool.

4. Select the *Unified Phantom Set* from the drop-down menu as shown in *Figure 19*.

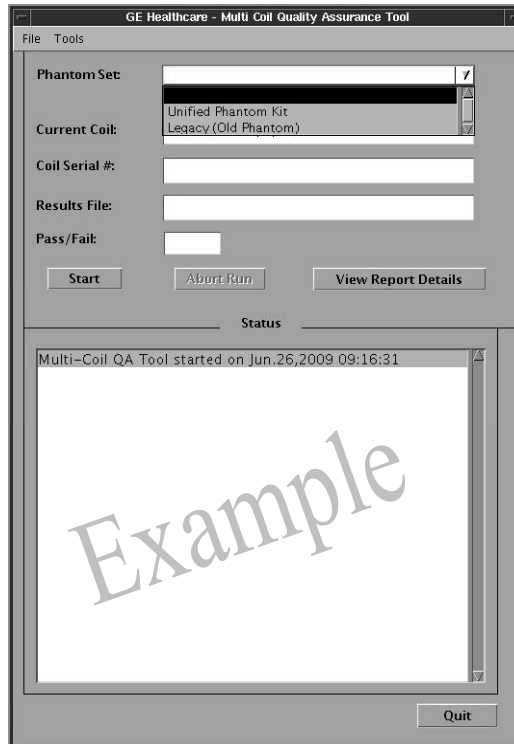


Figure 19

NOTE: For the Legacy Phantom Kit, please select LEGACY (old phantom).

5. The current coil field will be automatically filled in, based on the CoilID of the coil connected to the LPCA. Enter the serial number of the coil being tested in the *Coil Serial Number* field.
6. Click on [Start] to begin the automated test as shown in *Figure 20*. Depending on the number of test locations (complexity of the coil) the test may take from 3 to 20 minutes.

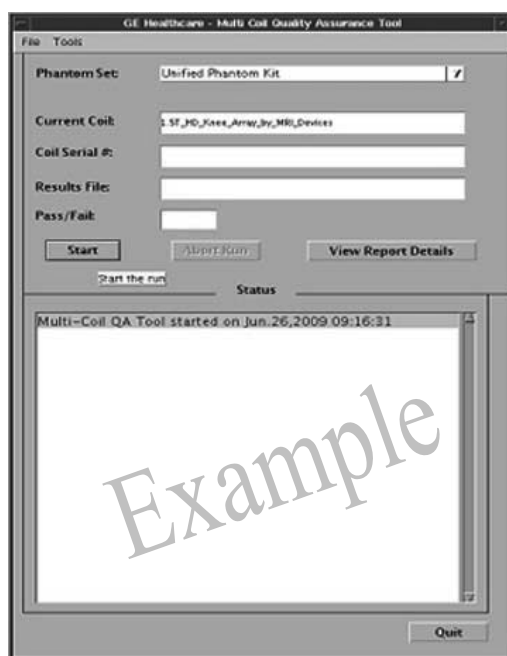


Figure 20

7. A confirmation pop-up displays, asking to continue. Click on [Yes]. See Figure 21.



Figure 21

8. Upon start-up, a message stating, **Phantom placement and coil landmarking are critical for repeatable results** will appear. If the landmark has been set correctly and there are no air bubbles in the phantom, click [Yes] to continue. See Figure 22.



Figure 22

NOTE: The Status window of the MCQA Tool GUI will continuously update to give information on what the tool is doing at any point in time. A time bar (Figure 23) will appear, showing approximate total test time, elapsed time and percent complete. Once the test has completed, the name of the Results file will be listed, along with the Pass/Fail status and the results of the test in the MCQA Tool GUI.

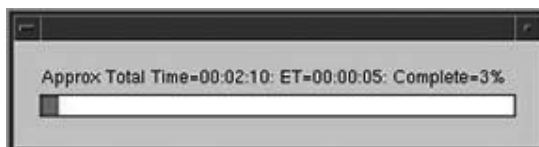


Figure 23

9. When the test is complete, test results display on the screen (Figure 24). The PASS/FAIL status shows PASS if all coil elements are functioning properly. If any coil elements display FAIL, call the GE Service Representative for coil repair.

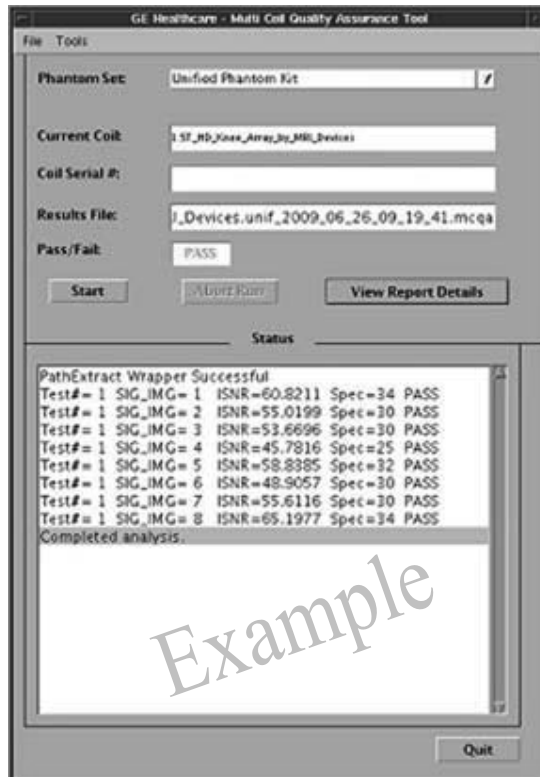


Figure 24

10. The MCQA Tool GUI displays **Fail** for reasons including, but not limited to:
 - a. Bad coil element
 - b. Incorrect phantom used for the test
 - c. Incorrect positioning/placement of the phantom
11. Click on [Quit] button to exit MCQA Tool

Viewing Report Results:

After running the Multi-Coil QA test, select [View Report Details] to review the results. To review past results, select *File, Open Results File* and select the desired coil results file. The Results Viewer will open as shown in Figure 25. The results file name and Pass/Fail Results shown on the tool GUI will also be listed across the top of the viewer.

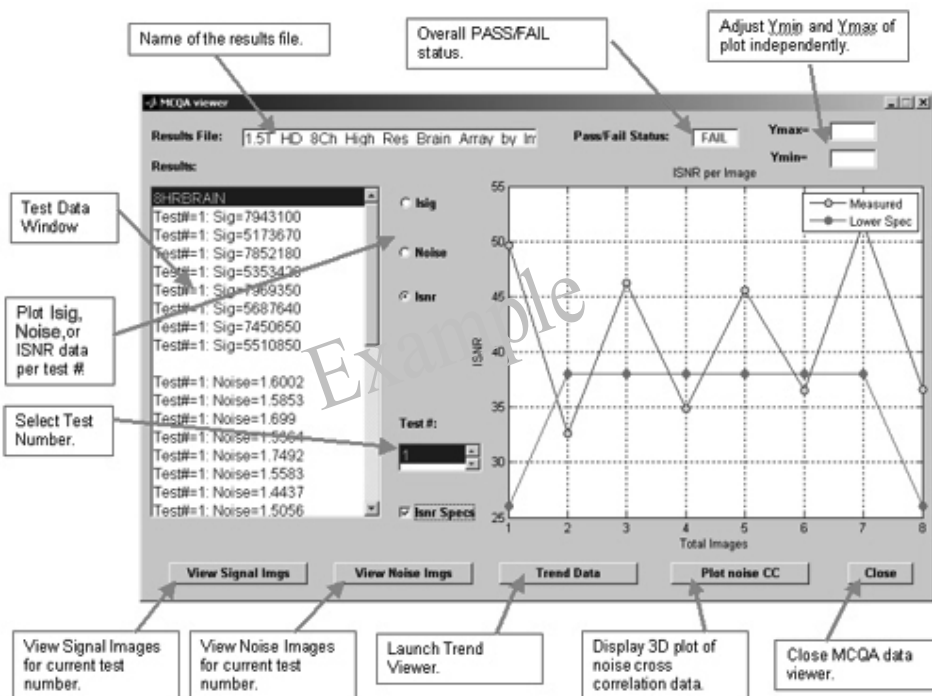


Figure 25

3-2-3 Test Explanation and Results Interpretation

The Premier III Phased Array CTL Spine Coil is a complex phased array. The Quality Assurance tool acquires data looking at the composite image (like the customer sees). Individual element performance is evaluated using the composite image.

Signal images are acquired using an SE sequence with CV saveinter set to 1 to save intermediate images into the image database. The scan is taken twice for a total of 42 images in the 8USCTL123service (or HDCTL 123serv) and 8USCTL456service (or HDCTL 456serv) modes.

The test employs a filtered version of the NEMA SNR measurement method. The filter takes out adverse effects of subtle ghosting on noise measured in difference.

3-3 External Cable Check

Visual check

Before removing the cable assembly from the coil, carefully check all coaxial connectors and pins on the coil connector. *Figure 26* shows the pin layouts of the coil connector. The connector of the CTL coil has 12 pins in Row B, eight pins in Row C and eight coaxial connectors in Row D. If there are any broken, deformed or recessed pins or coaxial connectors, replace the cable assembly.

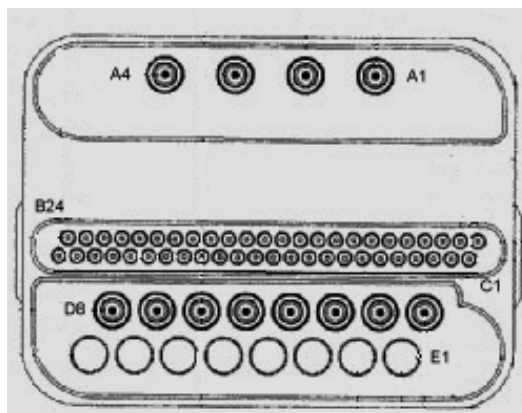


Figure 26: Pin layouts of Hypertronics connector.

Check for continuity of DC lines in the external cable

Step 1 – Remove the cable assembly from the coil as described in Section 5-1, External Cable Replacement.

Step 2 – There are six pins in Row B (see *Figure 26*) connecting to DC wires of the cable. *Table 3-3* shows these pins and corresponding DC wires. Use a Digital Volt Meter (DVM) to check the resistance between the pin in Row B and the corresponding DC wire (see *Figure 27*). The resistance should be $6.0 \pm 0.4\Omega$. If any resistance reading is not within this range, replace the cable assembly.

PINS AND CONNECTED DC WIRES – TABLE 3-3

Pin in Row B	10	11	12	13	14	15
DC wire color	Black	Brown	Red	Yellow	Green	Blue
DC bias of the system	MC1	MC2	MC3	MC4	MC5	MC6

Step 3 – Check if any pin from B10 to B15 is shorted to the ground. The outside conductor of a coaxial connector in Row D can be used as ground. If any pin is shorted to ground, replace the cable assembly.



Figure 27: Measure the resistance of the DC Line

Check for continuity of the preamp bias

The Pin 21 in row C is connected with eight RF cables. The system supplies the preamp bias voltage through Pin C21 and RF cables to the coil. Referring to *Figure 15*, find Pin 21 in Row C. Use a DVM to check resistance between Pin C21 and the central pin of the SMB connector at the end of each RF cable (see *Figure 28*). The resistance should be $6.0 \pm 0.4\Omega$. If any resistance reading is not within this range, replace the cable assembly.



Figure 28: Measure the resistance of the preamp bias line

3-4 PIN Diodes Check

This coil does not contain replaceable PIN diodes. No PIN diodes check is required.

3-5 Troubleshooting Tips

Symptom #1: The system reports a coil fault during prescan or does not recognize the coil connection to the system when selected in the software.

Probable Cause	Suggested Actions	Resolution
The coil connector has become disconnected from the system interface.	Check to make sure the coil connector is fully engaged.	Engage connector and try the scan again.
There is a DC short or open somewhere within the DC path inside the coil.	Perform Section 3-3, External Cable Check.	If there is no cable damage detected, replace the coil.
The output cable has a short or an open.	Perform Section 3-3, External Cable Check, to confirm.	Replace the coil cable assembly.
There is a problem with the DC bias from the system interface	Check to ensure the correct DC bias is being provided to the coil during transmit and receive per GE MRI System Service Manual. Try a similar 8-channel coil to see if the same problem occurs	Correct MRI system problem.
Coil ID issues.	From CSD, select Diagnostics, Manual Tests, SRI Functional Checks, and run the SRI Coil ID test. Compare the coil ID value found to that listed in the coil configuration.	If the value does not match, or no coil ID is found, replace the cable assembly.

Symptom #2: The coil does not pass the system level signal to noise check per Coil Imaging Performance Verification test or exhibits poor image quality on patient scans.

Probable Cause	Suggested Actions	Resolution
One or more of the coil channels has a high noise level.	Look at the uncombined images to determine which channel has the problem. Compare the noise standard deviation measurements between channels and system performance logs; they should be approximately the same.	Replace the coil.
One or more of the coil channels has low signal.	Look at the uncombined images to determine which channel has the problem. Compare signal mean measurements between channels and system performance logs.	Replace the coil.
Excessive ghosting is causing the noise standard deviation measurements to be artificially high.	Window and level the images down to look at the background for signs of ghosting. Try padding the phantom in the coil. Try turning off the cold heads to see if the ghosting is vibration-related.	If padding or changing the phantom position can minimize ghosting, then the problem is caused by excessive vibration elsewhere in the system. If the ghosting is not symptomatic of phantom positioning or padding, then replace the coil.
There is a problem with the DC bias from the system interface.	Check to ensure the correct DC bias is being provided to the coils during transmit and receive per GEMS System Service Manual.	Correct MRI system DC bias problem.

SECTION 4 – MAINTENANCE

4-1 Coil Care

WARNING!

Detach coil connector from the scanner before attempting to clean. Do not reattach after cleaning until the coil has dried completely. Having the coil attached to the system during cleaning or when it is wet may result in electrical shock.

CAUTION

4-1-1 Coil Care

The Premier III Phased Array CTL Spine Coil and patient comfort pads must be cleaned when needed and stored using the following procedure. Wipe with a cloth that has been dampened in a solution of 10 percent bleach and 90 percent tap water, or 30 percent isopropyl alcohol and 70 percent tap water.

Do not spray or pour any cleaning solution directly on the coil!

Store the coil in an air-conditioned scan room or equipment room.

Under no circumstances should the coil be placed into any type of sterilizer.

4-1-2 Storage Requirements

Store the coil in an air-conditioned scan room or equipment room.

To store the coil, a storage space exceeding the coil dimensions of 18.5" W x 44.7" L x 9.3" H (46.99cm W x 113.54cm L x 23.62cm H) is required.

SECTION 5 – REPLACEMENT

Simple removals that are clearly obvious are not described here.

Unless otherwise noted, the steps for re-assembly are simply the reverse order of the steps described for disassembly.

5-1 External Cable Replacement

Step 1 – Remove coil bottom cover by removing all screws around the perimeter and down the center of the base.

Step 2 – Remove the cable clamp that holds down the cable assembly (see *Figure 29*).

Step 3 – Disconnect the SMB plugs and DC plug on the cable end from the interface boards.

Step 4 – Remove the old cable assembly.

Step 5 – Attach the new cable assembly (see *Figure 30*) to the coil and make sure the cable strain relief matches the notch of the former (see *Figure 31*). The side of the cable connector with the “do not” label should face up (see *Figure 31*). If it doesn’t face up, rotate the strain relief. Make sure that the cable clamp holds the cable jacket, and then tighten the cable clamp with the screws. Connect all eight SMB plugs and the DC Plug to the corresponding receptacles on the interface board. The receptacles are marked CH1-8 to correspond with the labels on the cables. Do not route RF cables and DC wires above the bosses shown in *Figure 29* because the corresponding bosses on the bottom cover will squeeze the cables and DC wires when the bottom cover is closed.

Step 6 – Reattach the coil bottom cover.

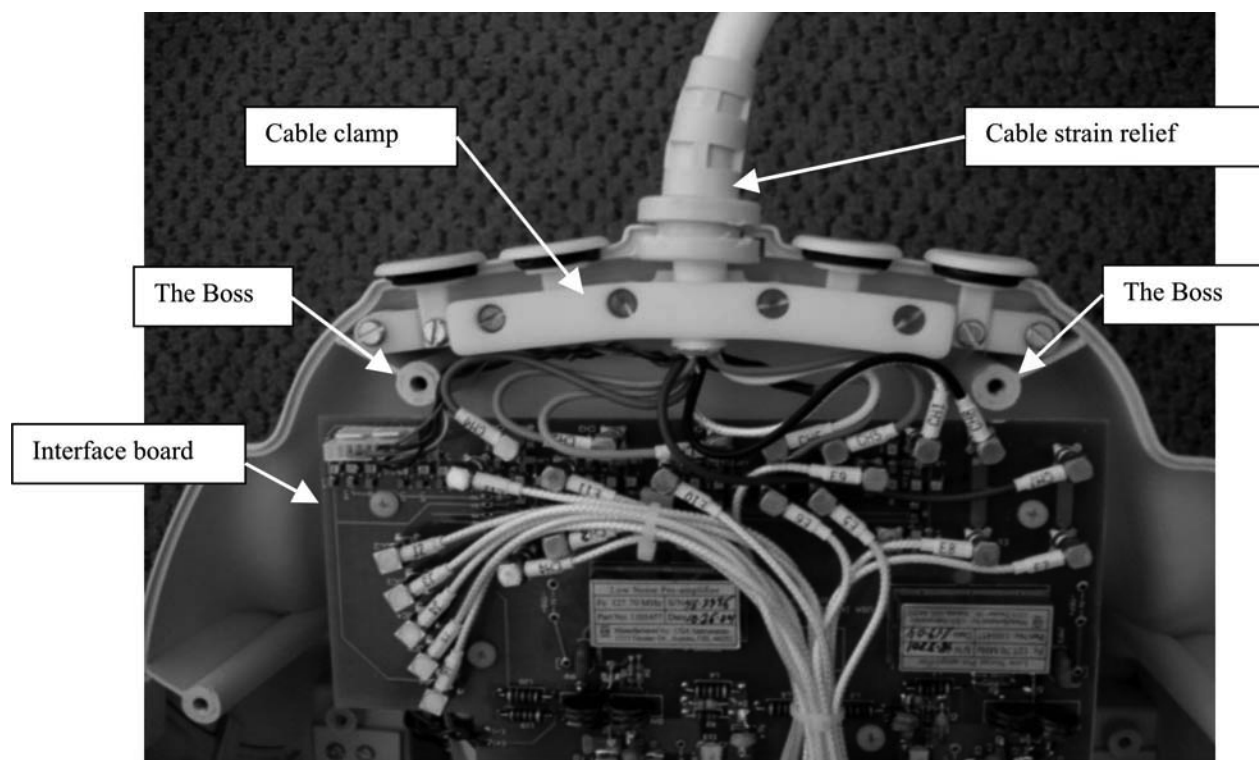


Figure 29: The Connection of the Cable Assembly and the Interface Board

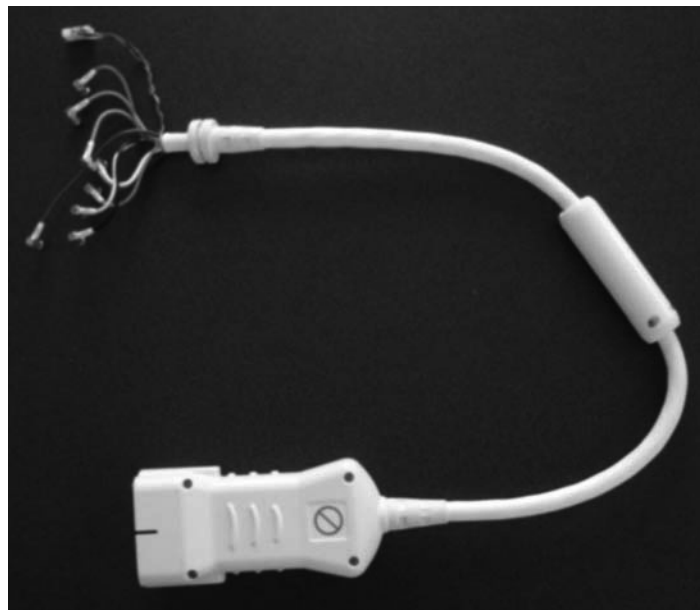


Figure 30: The cable assembly of the CTL coil

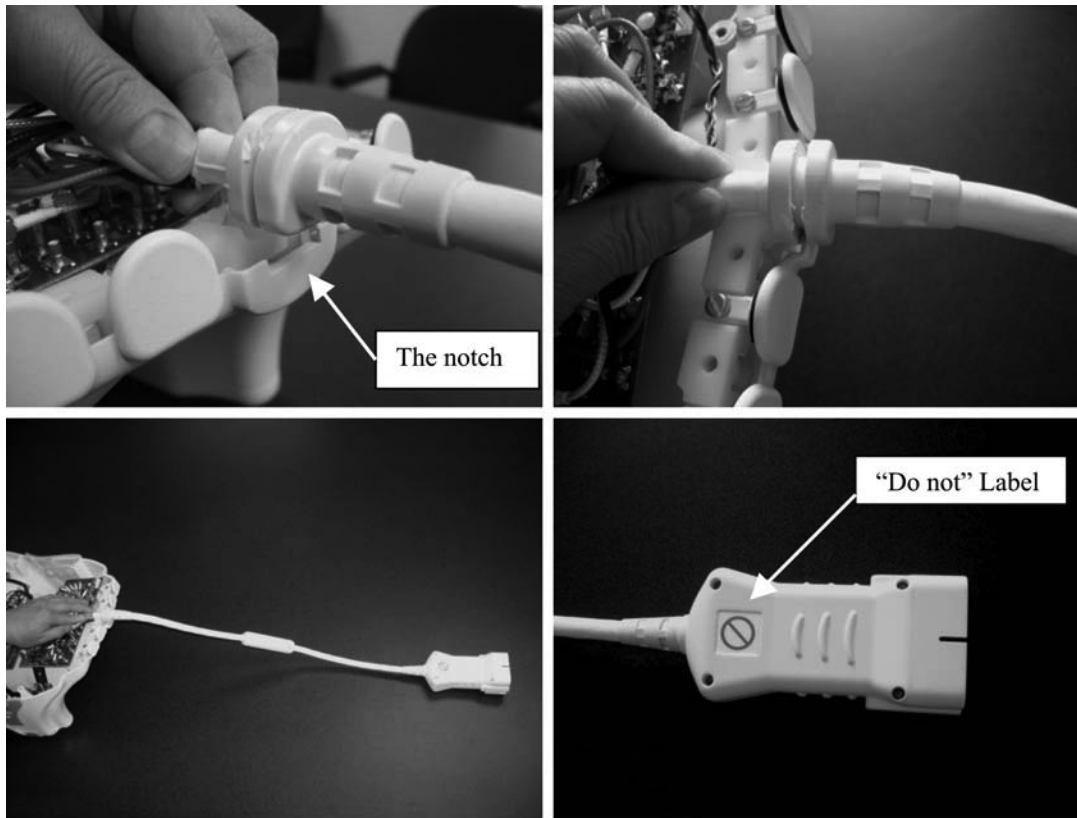


Figure 31: Install the cable assembly

SECTION 6 – RENEWAL PARTS

6-1 Field Replaceable Units

FIELD REPLACEABLE UNITS – TABLE 6-1

Part Name	GE Part #
Premier III Phased Array CTL Spine Coil (GE 3.0T HD CTL Array) w/ Cable	2413107-1
Premier III Phased Array CTL Spine Coil (GE 3.0T HD CTL Array) Cable	2413107-4
3T 8Ch CTL Phantom CT Section	2381682-9
3T 8Ch CTL Phantom TL Section	2381682-10

6-2 Other Replaceable Accessories

OTHER REPLACEABLE ACCESSORIES – TABLE 6-2

Part Name	GE Part #
Patient Comfort Pad	2413107-5
Ear Pad Set	2414538
Disposable Ear Pad Covers	2380634-8

REVISION HISTORY

Rev	Date	Author	Primary Reason for Change
1	Jan-05	Limin Feng	First Issue
2	08/20/08	Tricia Luthringer	Table 1-1: Changed p/n's for coil and Op guide. Section 1-1: Revised picture of coil. Section 1-3: changed p/n for Op guide. Section 3-2-2: Changed the Landmark information and step 5. Revised page header. Section 5-1: Revised Step 5 & 6. Revised tables 2-1-1 and 2-1-2.
3	19Apr10	D Seaman	Updated damage in transportation and language policy. Updated title page, all headers and footers, figure numbers. Updated photo of coil and included serial number label. Updated coil p/n in shipping list. Updated environmental requirements in table 1-4. Clarified Step 5 of section 5-1. Added metric measurements to storage requirements. Added new MCQA procedure for Unified phantom set per the Phantom Elimination Program